

The Geometry of Prime Vacuums: Legendre's Conjecture in Function Fields via Monodromy and Chebotarev Density

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Abstract

We prove the function field analogue of Legendre's Conjecture. Let $f(t) \in \mathbb{F}_q[t]$ be a squarefree monic polynomial of degree $d \geq 1$ with $\text{char}(\mathbb{F}_q) > 2d$. We show that for $q > (8d)^{2d+6}$, the short interval $\mathcal{I}_f = \{f(t)^2 + s(t) : \deg s \leq d\}$ contains at least one irreducible polynomial over $\mathbb{F}_q$; in fact, the number of such irreducibles is asymptotically $q^{d+1}/(2d)$. The proof constructs the universal polynomial family over the parameter space $\mathbb{A}^{d+1}$, establishes that its geometric monodromy group is the full symmetric group S_{2d} (via irreducibility, primitivity, and simple branching), and then applies the effective Chebotarev density theorem for higher-dimensional varieties. The topological error term is controlled by Katz's Betti number bounds.

1 Introduction

1.1 Background

The classical Legendre Conjecture posits that for every integer $n \geq 1$, there exists at least one prime in the interval $[n^2, (n+1)^2]$. This remains open over $\mathbb{Z}$; the best unconditional result (Baker–Harman–Pintz [3]) yields primes in $[x, x + x^{0.525}]$, which falls short of the $\mathcal{O}(x^{0.5})$ length of the Legendre interval.

Over the polynomial ring $\mathbb{F}_q[t]$, the geometry of algebraic varieties over finite fields provides the additional structure needed to resolve the problem. Our method follows the paradigm introduced by Bary-Soroker [4], who proved the function field analogue of Dirichlet's theorem on primes in arithmetic progressions using the monodromy of polynomial families and the Chebotarev density theorem. We adapt and extend this framework to the substantially different setting of *short intervals*, where the shift $s(t)$ perturbs only the lower-order coefficients of $f(t)^2$, introducing rigidity constraints that require a more delicate analysis of primitivity and branching.

1.2 Context within the Titan Project

This paper is part of a broader programme investigating the geometric and cohomological obstructions to classical prime number conjectures. In [5], we studied conductor rigidity phenomena for Frey curves associated to prime gaps, establishing that the Wiles paradigm encounters fundamental obstructions in the additive setting; this work provided the original motivation for seeking alternative geometric approaches to Legendre-type problems. In [6], we investigated

Landau’s fourth problem ($n^2 + 1$ primes) via conductor rigidity and Sato–Tate equidistribution for quadratic polynomial families—a setting closely analogous to the present one, where the quadratic structure $f(t)^2 + s(t)$ plays a similar role. In [7], we examined primes in arithmetic progressions through the lens of conductor rigidity, encountering the Bombieri–Vinogradov barrier; the monodromy and Chebotarev density framework employed in the present paper may be viewed as a function field resolution of the equidistribution difficulties identified there.

The present work differs from the preceding papers in a fundamental way: rather than analyzing conductor invariants of fixed L -functions, we construct a *universal polynomial family* and exploit the full symmetric monodromy of the associated covering space. This shift from local rigidity to global equidistribution is what enables a complete resolution in the function field setting.

1.3 Setup and main result

Definition 1.1 (Legendre Short Interval). Let $f(t) \in \mathbb{F}_q[t]$ be monic of degree $d \geq 1$. The *Legendre short interval* is

$$\mathcal{I}_f = \left\{ P_{\mathbf{a}}(t) = f(t)^2 + \sum_{i=0}^d a_i t^i \mid a_i \in \mathbb{F}_q \right\}.$$

The parameter space is $\mathcal{M} = \mathbb{A}_{\mathbb{F}_q}^{d+1}$ with coordinates $\mathbf{a} = (a_0, \dots, a_d)$. Every $P_{\mathbf{a}}$ is monic of degree $2d$, and $|\mathcal{I}_f| = q^{d+1}$.

Theorem 1.2 (Main Theorem). *Let $f(t) \in \mathbb{F}_q[t]$ be a squarefree monic polynomial of degree $d \geq 1$, and suppose $\text{char}(\mathbb{F}_q) > 2d$. Then the number of irreducible polynomials in $\mathcal{I}_f$ satisfies*

$$N_{\text{irr}} = \frac{q^{d+1}}{2d} + O_d\left(q^{d+\frac{1}{2}}\right). \quad (1)$$

In particular, $N_{\text{irr}} \geq 1$ whenever $q > (8d)^{2d+6}$.

Remark 1.3 (On the squarefree hypothesis). The assumption that f is squarefree is used in Proposition 2.6 to ensure that simple branching points exist on the discriminant locus. When f has repeated roots, $f(t)^2$ acquires roots of multiplicity ≥ 4 , and the analysis of local monodromy at such higher-order singularities requires versal deformations of A_k -singularities [1]. We expect Theorem 1.2 to hold without the squarefree condition, but we impose it here for clarity.

The proof occupies Sections 2–5 and proceeds in three steps:

- (i) Construct the universal polynomial family and prove its geometric monodromy group is S_{2d} (Section 2).
- (ii) Apply the effective Chebotarev density theorem for higher-dimensional parameter spaces (Section 3).
- (iii) Bound the topological error term via Katz’s Betti number estimates, complemented by the Grothendieck–Ogg–Shafarevich formula on one-dimensional slices (Section 4).

2 The Universal Family and Its Monodromy Group

2.1 The universal polynomial

Fix a squarefree monic polynomial $f(t) = t^d + c_{d-1}t^{d-1} + \dots + c_0 \in \mathbb{F}_q[t]$. Define the *universal polynomial*

$$F(t, \mathbf{a}) = f(t)^2 + a_0 + a_1 t + \dots + a_d t^d \in \mathbb{F}_q[\mathbf{a}][t], \quad (2)$$

a monic polynomial of degree $2d$ in t . Let $\mathcal{H} = \{F(t, \mathbf{a}) = 0\} \subset \mathcal{M} \times \mathbb{A}_t^1$, and $\pi : \mathcal{H} \rightarrow \mathcal{M}$ the projection. This is a finite morphism of degree $2d$.

The coefficient of t^k in F is:

$$[t^k]F = \begin{cases} [t^k]f(t)^2 & \text{if } d+1 \leq k \leq 2d, \\ [t^k]f(t)^2 + a_k & \text{if } 0 \leq k \leq d. \end{cases} \quad (3)$$

The top $d-1$ sub-leading coefficients (of $t^{2d-1}, \dots, t^{d+1}$) are fixed constants determined by f , while the bottom $d+1$ coefficients (of $t^d, \dots, t^0$) vary freely with $\mathbf{a}$. This *split structure* is crucial for the primitivity argument.

Definition 2.1. The *discriminant locus* is $\mathcal{D} = \{\mathbf{a} \in \mathcal{M} : \text{disc}_t(F(t, \mathbf{a})) = 0\}$. Over $\mathcal{M}^\circ = \mathcal{M} \setminus \mathcal{D}$, the morphism π is étale, defining the *geometric monodromy group*

$$G_{\text{geom}} = \text{Im} \left(\pi_1^{\text{ét}}(\mathcal{M}_{\mathbb{F}_q}^\circ) \rightarrow S_{2d} \right).$$

We establish $G_{\text{geom}} = S_{2d}$ in three steps.

2.2 Step 1: Irreducibility (Transitivity)

Proposition 2.2. *The polynomial $F(t, \mathbf{a})$ is irreducible over $\mathbb{F}_q(\mathbf{a})$.*

Proof. Write F as an element of $R[a_0]$ where $R = \mathbb{F}_q[t, a_1, \dots, a_d]$:

$$F = a_0 + (f(t)^2 + a_1 t + \dots + a_d t^d).$$

Since F is *degree one* in a_0 with coefficients in the integral domain R , it is irreducible in $R[a_0] = \mathbb{F}_q[t, a_0, \dots, a_d]$.

This ring is a UFD (polynomial ring over a field). By Gauss's lemma, if F were reducible in $\mathbb{F}_q(\mathbf{a})[t]$, clearing denominators would yield a nontrivial factorization in $\mathbb{F}_q[\mathbf{a}, t]$, contradicting the above. $\square$

Corollary 2.3. G_{geom} acts transitively on the $2d$ roots of F .

Proof. Irreducibility over $\mathbb{F}_q(\mathbf{a})$ implies the generic fiber of $\mathcal{H}^\circ \rightarrow \mathcal{M}^\circ$ is connected, equivalent to transitivity ([22, Exp. V, Prop. 8.2]). $\square$

2.3 Step 2: Primitivity

A transitive subgroup $G \leq S_n$ is *primitive* if it preserves no nontrivial block partition of $\{1, \dots, n\}$. For polynomial coverings, imprimitivity is equivalent to a functional decomposition $F(t) = g(h(t))$ with $1 < \deg h < n$ ([16, 24]).

Proposition 2.4. G_{geom} is primitive.

Proof. Suppose for contradiction that there exist monic polynomials $h \in \mathbb{F}_q(\mathbf{a})[t]$ of degree m and $g \in \mathbb{F}_q(\mathbf{a})[x]$ of degree $n = 2d/m$ (with $1 < m < 2d$, $m \mid 2d$) such that

$$F(t, \mathbf{a}) = g(h(t, \mathbf{a}), \mathbf{a}). \quad (4)$$

Since $m \mid 2d$ and $m < 2d$, we have $m \leq d$ and $n \geq 2$.

Normalization. Replacing $h(t)$ by $h(t) - h(0)$ and $g(x)$ by $g(x + h(0))$ preserves (4) while achieving

$$h(0) = 0. \quad (5)$$

Write $h(t) = t^m + h_{m-1}t^{m-1} + \dots + h_1 t$ (with $h_i \in \overline{\mathbb{F}}_q(\mathbf{a})$, no constant term), and $g(x) = x^n + g_{n-1}x^{n-1} + \dots + g_0$.

Coefficient analysis. For $1 \leq j \leq m-1$, the coefficient $[t^{2d-j}]$ in the expansion of $g(h(t))$ depends only on $h(t)^n$ at this degree (since $g_{n-1}h(t)^{n-1}$ contributes at degree $\leq 2d-m$, and $2d-j > 2d-m$ for $j < m$). Specifically:

$$[t^{2d-j}]g(h(t)) = n h_{m-j} + \Phi_j(h_{m-1}, \dots, h_{m-j+1})$$

for a universal polynomial Φ_j (with $\Phi_1 = 0$).

Since $m \leq d$, we have $2d-j \geq 2d-(m-1) \geq d+1$ for all $1 \leq j \leq m-1$. By the split structure (3), $[t^{2d-j}]F$ is a *fixed constant* in $\mathbb{F}_q$ for these degrees.

Equating inductively: for $j = 1$, $n \cdot h_{m-1} = 2c_{d-1}$, so $h_{m-1} = 2c_{d-1}/n \in \mathbb{F}_q$ (using $\gcd(n, \text{char}) = 1$ since $n \leq 2d < p$). For $j = 2, \dots, m-1$, each step determines h_{m-j} as a fixed element of $\mathbb{F}_q$ in terms of previously determined coefficients. Thus $h_{m-1}, \dots, h_1 \in \mathbb{F}_q$ are all fixed constants. Combined with the normalization $h_0 = 0$, the polynomial $h(t) \in \mathbb{F}_q[t]$ is entirely independent of $\mathbf{a}$.

Contradiction. Since $h \in \mathbb{F}_q[t]$ is fixed, $F(t, \mathbf{a}) = g(h(t), \mathbf{a})$. The coefficients of F at degrees $0 \leq k \leq d$ are $[t^k]f(t)^2 + a_k$, providing $d+1$ independent free parameters $(a_0, \dots, a_d)$. But g has only $n = 2d/m$ free coefficients $(g_0, \dots, g_{n-1})$, and $n \leq d < d+1$. The polynomial map $(g_0, \dots, g_{n-1}) \mapsto ([t^0]g(h(t)), \dots, [t^d]g(h(t)))$ goes from $\mathbb{A}^n$ to $\mathbb{A}^{d+1}$ with $n < d+1$, hence is not surjective. This contradicts the fact that the left-hand side ranges over all of $\mathbb{A}^{d+1}$ as $\mathbf{a}$ varies. $\square$

Remark 2.5. When $d = 1$, the group S_2 has no proper transitive subgroup, so primitivity is automatic.

2.4 Step 3: Transpositions from simple branching

Proposition 2.6. *Assume $f(t)$ is squarefree and $\text{char}(\mathbb{F}_q) > 2d$. Then G_{geom} contains a transposition.*

Proof. We exhibit smooth points of $\mathcal{D}$ where exactly two roots of F coalesce.

Setup. Since f is squarefree, it has d distinct roots $\alpha_1, \dots, \alpha_d \in \overline{\mathbb{F}}_q$. At $\mathbf{a} = \mathbf{0}$, the polynomial $F(t, \mathbf{0}) = f(t)^2$ has each α_j as a double root.

Independent unfolding via Vandermonde. Consider the evaluation map $\varphi : \overline{\mathbb{F}}_q^d \rightarrow \overline{\mathbb{F}}_q^d$ defined by

$$\varphi(a_0, \dots, a_{d-1}) = (P(\alpha_1), P(\alpha_2), \dots, P(\alpha_d)),$$

where $P(t) = a_0 + a_1 t + \dots + a_{d-1} t^{d-1}$. The Jacobian of φ is the Vandermonde matrix $V(\alpha_1, \dots, \alpha_d)$, whose determinant $\prod_{i < j} (\alpha_j - \alpha_i)$ is nonzero since the α_j are distinct. Therefore φ is an isomorphism, and the d parameters $(a_0, \dots, a_{d-1})$ independently control the perturbation at each double root.

Concretely, near α_j and $\mathbf{a} = \mathbf{0}$, the local equation of F in the coordinate $u = t - \alpha_j$ is

$$F(\alpha_j + u, \mathbf{a}) = f'(\alpha_j)^2 u^2 + P(\alpha_j) + O(u \cdot \mathbf{a}) + O(u^3).$$

Since f is squarefree, $f'(\alpha_j) \neq 0$ (using $\text{char}(\mathbb{F}_q) > 2d \geq \deg f' + 1$). The double root at $u = 0$ splits into two simple roots when $P(\alpha_j) \neq 0$, or coalesces when $P(\alpha_j) = 0$. The invertibility of φ means these d conditions are independent.

Identifying simple nodes. The discriminant locus $\mathcal{D}$ near $\mathbf{a} = \mathbf{0}$ has d smooth branches $\mathcal{D}_j = \{\mathbf{a} : P(\alpha_j) = 0\}$ for $j = 1, \dots, d$, intersecting transversally at the origin. A general point on $\mathcal{D}_j$ (away from the other branches) corresponds to a polynomial $P_{\mathbf{a}}$ with exactly one double root (near α_j) and $2d-2$ simple roots.

Local monodromy. By the Picard–Lefschetz formula ([23, Exp. XV, §3]; [13, §I.6]), an étale loop around the smooth locus of $\mathcal{D}_j$ transposes the two coalescing sheets and fixes all others. This is a transposition in G_{geom} . $\square$

Remark 2.7. When f is not squarefree, double roots of f become fourfold roots of f^2 , and local monodromy may produce products of transpositions rather than a single transposition. Extracting an individual transposition then requires A_k -singularity analysis [1]. We leave this extension to future work.

2.5 Conclusion: the full symmetric group

Theorem 2.8. *Let $f \in \mathbb{F}_q[t]$ be squarefree monic of degree $d \geq 1$ with $\text{char}(\mathbb{F}_q) > 2d$. Then $G_{\text{geom}} = S_{2d}$.*

Proof. G_{geom} is transitive (Corollary 2.3), primitive (Proposition 2.4), and contains a transposition (Proposition 2.6). By Jordan’s theorem ([10, Thm. 3.3E]), a *primitive* subgroup of S_n containing a transposition equals S_n . Hence $G_{\text{geom}} = S_{2d}$. $\square$

Remark 2.9. The hypothesis of *primitivity* is essential. The weaker statement “transitive + contains a transposition $\Rightarrow S_n$ ” is false: the dihedral group $D_4 \leq S_4$ is transitive and contains transpositions, but $D_4 \neq S_4$.

3 Counting Irreducibles via the Chebotarev Density Theorem

3.1 Frobenius elements and factorization type

For $\mathbf{a} \in \mathcal{M}^\circ(\mathbb{F}_q)$, the Frobenius element $\text{Frob}_{\mathbf{a}}$ is a well-defined conjugacy class in S_{2d} whose cycle type $(d_1, \dots, d_k)$ encodes the degrees of the irreducible factors of $P_{\mathbf{a}}$ over $\mathbb{F}_q$. Thus $P_{\mathbf{a}}$ is irreducible $\Leftrightarrow \text{Frob}_{\mathbf{a}}$ is a $2d$ -cycle.

Lemma 3.1. *The proportion of $2d$ -cycles in S_{2d} is $1/(2d)$.*

Proof. There are $(2d - 1)!$ such cycles, so the proportion is $(2d - 1)!/(2d)! = 1/(2d)$. $\square$

3.2 The effective Chebotarev theorem for higher-dimensional varieties

We use the following result, which combines the Grothendieck–Lefschetz trace formula with Deligne’s purity theorem [9]. The formulation via the *permutation sheaf* (rather than the Galois closure) avoids factorial-sized constants.

Theorem 3.2 (Effective Chebotarev; cf. [13, Thm. 9.2.6], [11, §9.7]). *Let $U/\mathbb{F}_q$ be a smooth geometrically connected affine variety of dimension n , and let $\pi : \mathcal{Y} \rightarrow U$ be a finite étale cover of degree N with connected geometric generic fiber, so that the monodromy group $G \leq S_N$ is transitive. Let $\mathcal{F} = \pi_* \overline{\mathbb{Q}}_\ell$ be the associated permutation sheaf of rank N on U , and let $C \subset G$ be a conjugacy class. Then*

$$\left| \#\{u \in U(\mathbb{F}_q) : \text{Frob}_u \in C\} - \frac{|C|}{|G|} q^n \right| \leq B(\mathcal{F}) \cdot q^{n-1/2} + O_{N,n}(q^{n-1}), \quad (6)$$

where $B(\mathcal{F}) = \sum_{i=0}^{2n} \dim H_c^i(U_{\overline{\mathbb{F}}_q}, \mathcal{F})$ is the total Betti number of the sheaf $\mathcal{F}$ on U (not the Betti number of the covering space).

Remark 3.3. The decisive advantage of formulating the error in terms of $B(\mathcal{F})$ rather than the Betti numbers of the Galois closure is that the former avoids factors of $|G|$ (which is $(2d)!$ for $G = S_{2d}$). By [14, Thm. 12], for the complement U of a hypersurface of degree δ in $\mathbb{A}^n$:

$$B(\mathcal{F}) \leq N \cdot (2\delta + 1)^n. \quad (7)$$

In our setting, $N = 2d$, $n = d + 1$, and δ is the *total degree* of $\text{disc}_t(F)$ as a polynomial in $\mathbf{a}$. Since $F(t, \mathbf{a})$ is monic of degree $2d$ in t and each coefficient of F is at most degree 1 in $\mathbf{a}$, the discriminant — being the resultant of F and F' , hence a determinant of size $(4d - 1)$ whose entries are coefficients of F and F' — has total degree at most

$$\delta = 2(2d) - 2 = 4d - 2 \quad (8)$$

in $\mathbf{a}$ (this is the classical bound for the total degree of the discriminant of a monic polynomial of degree N in its coefficients; it should *not* be confused with the *weighted* degree $N(N - 1) = 2d(2d - 1)$, which assigns weight $N - i$ to the i -th coefficient).

Hence $2\delta + 1 = 8d - 3$ and:

$$B(\mathcal{F}) \leq 2d \cdot (8d - 3)^{d+1} \leq 2d \cdot (8d)^{d+1}. \quad (9)$$

The essential point: $B(\mathcal{F})$ depends on d alone, not on q .

4 Topological Complexity and the GOS Formula

While the Katz bound (9) suffices for the asymptotic result, the Grothendieck–Ogg–Shafarevich (GOS) formula provides sharper bounds on one-dimensional slices and illuminates the geometric mechanism.

4.1 The GOS formula on a generic pencil

Let $\ell \cong \mathbb{A}^1 \hookrightarrow \mathcal{M}$ be a general affine line. After restricting and compactifying, we obtain a lisse $\overline{\mathbb{Q}}_\ell$ -sheaf $\mathcal{F}_\ell$ of rank $r = 2d - 1$ (the reduced permutation representation) on an open $U_\ell \subset \mathbb{P}_{\mathbb{F}_q}^1$. The GOS formula [12, 17, 20] states:

$$\chi(\mathbb{P}_{\mathbb{F}_q}^1, j_! \mathcal{F}_\ell) = 2r - \sum_{v \in \mathbb{P}^1 \setminus U_\ell} (\text{drop}_v + \text{Sw}_v). \quad (10)$$

4.2 Tameness kills wild ramification

Under $\text{char}(\mathbb{F}_q) = p > 2d$:

At finite bad points v : the inertia group I_v acts through a cyclic group of order dividing $2d$, which is coprime to p . Hence $\text{Sw}_v(\mathcal{F}_\ell) = 0$ ([20, IV, §2, Cor. 1]).

At $v = \infty$: the ramification indices divide $|S_{2d}| = (2d)!$. Since $p > 2d$, every integer from 1 to $2d$ is coprime to p , so the wild inertia subgroup (a p -group) acts trivially on the $2d$ roots. Thus $\text{Sw}_\infty = 0$.

4.3 One-dimensional Euler characteristic bound

Since all Swan conductors vanish, (10) gives $\chi = 2r - \sum_v \text{drop}_v$ with each $0 \leq \text{drop}_v \leq r = 2d - 1$. By (8), the number of finite bad fibers on a generic pencil is at most $D = 4d - 2$, plus one point at ∞ . Hence:

$$|\chi(\mathbb{P}_{\mathbb{F}_q}^1, j_! \mathcal{F}_\ell)| \leq (D + 2) \cdot r \leq 4d(2d - 1) \leq 8d^2. \quad (11)$$

This provides a tight polynomial bound on one-dimensional slices. The passage to the full $(d + 1)$ -dimensional parameter space uses the Katz bound (9) as stated in Remark 3.3.

5 Proof of the Main Theorem

Proof of Theorem 1.2. Step 1: Monodromy. By Theorem 2.8, $G_{\text{geom}} = S_{2d}$.

Step 2: Chebotarev count. Apply Theorem 3.2 with $U = \mathcal{M}^\circ$, $n = d + 1$, $N = 2d$, $C =$ conjugacy class of $2d$ -cycles. By Lemma 3.1, $|C|/|G| = 1/(2d)$. Using the Betti bound (9):

$$N_{\text{irr}}^\circ := \#\{\mathbf{a} \in \mathcal{M}^\circ(\mathbb{F}_q) : P_{\mathbf{a}} \text{ irreducible}\} = \frac{q^{d+1}}{2d} + O_d(q^{d+1/2}). \quad (12)$$

Step 3: Discriminant locus. $\mathcal{D}$ is a hypersurface of degree $\leq 4d - 2$ in $\mathbb{A}^{d+1}$ (see (8)), so $|\mathcal{D}(\mathbb{F}_q)| \leq (4d - 2)q^d + O_d(q^{d-1/2})$ by Lang–Weil [15]. On $\mathcal{D}$, $P_{\mathbf{a}}$ has repeated roots, hence is reducible (since irreducible implies separable when $\text{char} > \deg$). Thus $N_{\text{irr}} = N_{\text{irr}}^\circ$.

Step 4: Explicit positivity. Write

$$N_{\text{irr}} \geq \frac{q^{d+1}}{2d} - B(\mathcal{F}) \cdot q^{d+1/2} - C'_d \cdot q^d, \quad (13)$$

where $B(\mathcal{F}) \leq 2d(8d)^{d+1}$ (from (9)) and $C'_d \leq 4d - 2$ absorbs the Lang–Weil term. For $q \geq 2$, the q^d term is dominated by $q^{d+1/2}$, so:

$$N_{\text{irr}} \geq q^{d+1/2} \left(\frac{q^{1/2}}{2d} - B(\mathcal{F}) - C'_d \right).$$

Since $C'_d \leq 4d \leq B(\mathcal{F})$ for $d \geq 1$, we may take $A_d = 2B(\mathcal{F}) \leq 4d(8d)^{d+1}$. Then $N_{\text{irr}} > 0$ when $q^{1/2} > 2d \cdot A_d$, i.e.,

$$q > (2d \cdot A_d)^2 = (8d^2(8d)^{d+1})^2 = 64d^4(8d)^{2d+2}.$$

Since $64d^4 \leq (8d)^4$ for $d \geq 1$, this yields:

$$q > (8d)^{2d+6}. \quad (14)$$

□

Remark 5.1 (On the threshold). The bound (14) is far from optimal; it is driven by the worst-case Katz estimate for the Betti numbers. The corrected discriminant degree $\delta = 4d - 2$ (as opposed to the weighted degree $2d(2d - 1)$ erroneously used in earlier versions) yields a significant improvement: the base of the exponential is $8d$ rather than $\sim 4d^2$. Numerical evidence (Section 6) strongly suggests that $N_{\text{irr}} \geq 1$ holds for *all* q and d with $\text{char}(\mathbb{F}_q) > 2d$. An approach via sieve methods or the large sieve inequality for function fields (as in [2]) could potentially yield much sharper bounds.

6 Numerical Verification

Example 6.1 ($d = 1$, $f(t) = t$). $\mathcal{I}_f$ is the set of all monic quadratics over $\mathbb{F}_q$. The exact count of irreducibles is $(q^2 - q)/2$, matching $q^2/2 + O(q)$.

Data. $q = 5$: $N_{\text{irr}} = 10$, predicted 12.5. $q = 7$: $N_{\text{irr}} = 21$, predicted 24.5. $q = 11$: $N_{\text{irr}} = 55$, predicted 60.5.

Example 6.2 ($d = 2$, $f(t) = t^2 + 1$, $q = 11$). f is squarefree (irreducible over $\mathbb{F}_{11}$). $|\mathcal{I}_f| = 11^3 = 1331$; predicted $\lfloor 1331/4 \rfloor = 332$; direct computation gives $N_{\text{irr}} = 330$.

Example 6.3 ($d = 1$, $f(t) = t$, $q = 3$). $\text{char} = 3 > 2 = 2d$, so the hypotheses hold. The explicit asymptotic threshold is astronomically large, yet $N_{\text{irr}} = 3 > 0$. This illustrates that the theorem's bound is sufficient but vastly non-sharp.

7 Discussion

7.1 The topological obstruction perspective

The proof reduces to one principle: *a connected covering with full symmetric monodromy cannot have all rational fibers reducible*. S_{2d} is the global symmetry; Chebotarev translates it into equidistribution of Frobenius elements; the Katz/GOS bounds ensure the error is $O(q^{-1/2})$ relative to the main term.

We emphasize that the GOS formula does *not* produce the contradiction directly (as was incorrectly attempted in an earlier version via “Swan conductor explosion”). Swan conductors are local invariants of a single sheaf and are always finite; the global factorization pattern is an arithmetic property of the ensemble of rational fibers, governed by equidistribution.

7.2 Relation to prior work

Bary-Soroker [4] proved the function field Dirichlet theorem by the same monodromy paradigm. Our contribution is the adaptation to short intervals, which differs in two respects: (a) the parameter space is $(d+1)$ -dimensional (requiring the higher-dimensional Chebotarev theorem), and (b) the rigid leading-term structure of $f(t)^2 + s(t)$ demands a non-trivial primitivity analysis (Proposition 2.4).

7.3 Relation to the classical Legendre Conjecture

The function field proof succeeds because the Riemann Hypothesis is a theorem (Weil, Deligne) and the étale fundamental group provides a computable monodromy group. Over $\mathbb{Z}$, both ingredients are absent or conjectural.

7.4 Open questions

- (1) *Removing the squarefree hypothesis*. See Remark 2.7.
- (2) *Uniformity in d* . Does $N_{\text{irr}} \geq 1$ for all q and d with $\text{char} > 2d$?
- (3) *Shorter intervals*. Minimize e such that $\{f^2 + s : \deg s \leq e\}$ contains an irreducible.
- (4) *Higher powers*. For $k \geq 3$, study $[f^k, (f+1)^k]$.

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